

**Chapter 2:**

a) Relationship between module and Session

The relationship is **one to many**, this means that one module can have zero or many sessions and a session must have one module

b) Relationship between session and schedule

The relationship is one to many, this means a session can have 0 or more schedules, but a schedule must have one session.

c) Relationship between schedule and student

The relationship is one to many, this means that a student can have at least one session and one schedule can be scheduled by one student

**Chapter 3:**

a)

| Table   | Primary key | Foreign key |
|---------|-------------|-------------|
| Student | stuNum      | N/A         |
| Staff   | StaffNum    | StaffNum    |

b)

| Table   | Yes/No | Explanation                                         |
|---------|--------|-----------------------------------------------------|
| Student | Yes    | There is no repetition of rows, each row is unique. |
| Staff   | Yes    | Each row is unique there is no repetition.          |

c)

| Table   | Yes/No | Explanation                                                   |
|---------|--------|---------------------------------------------------------------|
| Student | Yes    | There is no FK for student table.                             |
| Staff   | No     | The FK for Mpumelelo is empty or is null and it must be null. |

**Chapter 4:**

a)  $\pi$  studentNumber (Student)

| StuNum   |
|----------|
| 22405887 |
| 22365169 |

|          |
|----------|
| 22478172 |
| 22233435 |
| 22692868 |
| 22345427 |

b)  $\pi$  studentNumber, deptID( $\sigma$  STUDENT.staffNum = STAFF.staffNum (STUDENT  $\times$  STAFF))

| studentNum | deptID |
|------------|--------|
| 22365169   | 2119   |
| 22233435   | 2119   |
| 22478172   | 1871   |
| 22345427   | 2034   |

## Chapter 5:

### A)

#### Relationship between salesperson and Sale

The relationship is one to many relation (1:N) meaning that one salesperson can have one or more sales and one Sale can be done/completed by only one salesperson

#### Relationship between Sale and Product

The relationship is one to many this means that a sale must have one product and a product can have zero or more sales.

#### Relationship between Sale and Customer

The relationship is one to many then this means that a customer can be collected to one or more sales and one sale must have one customer.

#### Relationship between Product and Category

The relationship is one to many, this means that one product has only one productCategory but one productCategory can have 0 or more products.

#### Relationship on product itself (Recursive Relationship)

One salesperson can manage 0 or 1 salesPerson but one salesperson can be managed by one salesperson.

**B)** Weak because sale consists of primary key that is composite it consists of SPNumb,CNum,PCode and SaleID, so this means that Sale entity depends on other entities.

**C)** Strong because Sale consists of only one primary key and this shows that the Sale is independent and it does not depend on any entity.